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Article

NMR Relaxometry and Diffusometry Analysis of Dynamics in Ionic Liquids and Ionogels for Use in Lithium-Ion Batteries

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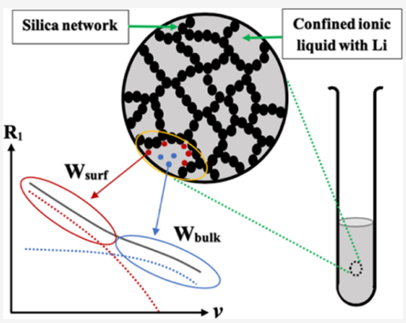

ABSTRACT: We have investigated the charge transport dynamics of a novel solid-like electrolyte material based on mixtures of the ionic liquid (IL) 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([BMIM] TFSI) and various concentrations of lithium salt bis(trifluoromethylsulfonyl)imide (LiTFSI) confined within a SiO_2 matrix, prepared via a sol−gel method. The translational diffusion coefficients of $BMIM^+$, $TFSI^-$, and Li^+ in ILs and confined ILs (ionogels, IGs) with different concentrations of lithium salt have been measured at variable temperatures, covering the 20−100 °C range, using nuclear magnetic resonance (NMR) pulsed field gradient diffusion spectroscopy. The mobility of $BMIM^+$, $TFSI^-$, and Li^+ was found to increase with the [BMIM] TFSI/LiTFSI ratio, exhibiting an almost liquid-like mobility in IGs. Additionally, the effect of confinement on IL rotational dynamics has been analyzed by measuring ^1H, ^{19}F, and ^7Li spin−lattice relaxation rate dispersions of IGs at different temperatures, using fast field-cycling NMR relaxometry. The analysis of the experimental data was performed assuming the existence of two fractions of the liquid: a bulk fraction (at least several ionic radii from the silica particles) and a surface fraction (close to the silica particles) and using two different models based on translational and rotational diffusion and reorientation mediated by translational displacements. The existence and weighting of these two fractions of ions were obtained from the direct diffusion measurements. The results show that the ion dynamics slowed only modestly under confinement, which evidences that IGs preserve IL transport properties, and this behavior is an encouraging indication for using IGs as a solid electrolyte for Li^+ batteries.

■ INTRODUCTION

Commercial lithium-ion (Li^+) batteries power an extraordinary range of devices from consumer electronics to the electrification of transportation and grid storage. While the energy and power considerations for the Li^+ batteries that power these devices vary widely, a common concern is that of safety as the liquid carbonate electrolyte solvents, which are widely used in these batteries, are flammable. For these and other reasons, there is widespread interest in the use of solid electrolytes whose high temperatures and chemical stability readily satisfy the safety requirements. The research described here is directed at determining the fundamental transport properties of a stable electrolyte for Li^+ batteries.

A key feature in the research described here is the use of an ionic liquid (IL). These materials are considered to be molten salts with melting temperatures below 100 °C. Numerous IL systems are liquid at room temperature and below. ILs have many useful properties, including low vapor pressure, a broad liquid-state temperature window, high chemical and thermal stability, wide electrochemical voltage window, nonflammability, high ionic conductivity, and a broad polar to non-polar solubility range.[1−3] These unique properties of ILs make them particularly promising candidates as an electrolyte for high-performance batteries, when partnered with an appropriate Li salt.

Solid-state electrolytes have been explored increasingly for use in Li^+ batteries because of their ability to address the safety issues, packaging constraints, and volumetric design concerns associated with liquid electrolytes.[3−6] However, solid electrolytes typically exhibit resistive interfaces and an overall low ionic conductivity.[3−7] Recently, ionogels (IGs), pseudo-solid-state electrolytes consisting of an IL electrolyte confined in a mesoporous inorganic matrix, have attracted interest because of their high ionic conductivity, stability, and solution processability, which ensures a well-wetted electrode/electrolyte interface. Macroscopically, the IG behaves as a solid, but at the nanoscale, a liquid-state is maintained.[3−10] Structural and dynamical properties of these confined ILs have been investigated using diverse experimental techniques such as quasi-elastic neutron scattering (QENS), solid-state magic angle spinning nuclear magnetic resonance (MAS NMR) spectroscopy, pulsed field gradient (PFG) NMR method, Raman spectroscopy, impedance spectroscopy, and fast field-

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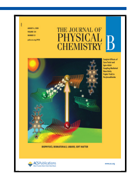

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cycling NMR (FFC-NMR) relaxometry.[3−9,11−17] Because of the complexity of the trapped liquid, these techniques are yet to fully address the nature of the ion interaction and movement. Understanding how the ions behave inside the matrix is paramount for increasing the ionic conductivity and the Li^+ transference number. The literature for these characteristics is lacking in several key areas including, but not limited to, variations in the electrolyte's Li^+ concentration, the matrix composition, and ion diffusion under different conditions.

Most of the NMR studies about IGs have been focused on the dynamics of the cation of the confined IL (^{1}H NMR),[11,12,15,16] and only few of them have studied both cation and anion dynamics (^{1}H and ^{19}F NMR).[14,17] However, all these works have shown that the confinement causes a slight slowing-down of the ion dynamics. These results are consistent with the fact that the ionic conductivity of the IG is usually smaller compared to the corresponding value for the pure IL.[16] Nevertheless, none of these works has studied the effect of different concentrations of lithium salt on ILs and IGs. To the best of our knowledge, this is the first study that combines the PFG-NMR diffusometry and FFC-NMR relaxometry techniques to elucidate the dynamics (translational and rotational diffusion) of ILs and IGs with variation of lithium content and temperatures. An additional motivation for this undertaking concerns the high likelihood that rotational and translational dynamics are coupled in these highly correlated systems.

In this paper, we report experimental data on an IG in which the IL, 1-butyl-3-methylimidazolium bis-(trifluoromethylsulfonyl)imide ([BMIM] TFSI), together with the lithium salt, lithium bis(trifluoromethylsulfonyl)imide (LiTFSI), are confined within a SiO_2 matrix. The objective of the present study is to understand the confinement effects on the translational and rotational motion of the ions of these IGs as compared to IL electrolytes. With this purpose in mind, translational diffusion coefficients of the three ions ($BMIM^+$, $TFSI^-$, and Li^+) and a multinuclear (^{1}H, ^{19}F, and ^{7}Li) variable temperature, and variable frequency study of the spin−lattice relaxation rate (R_1), for both ILs and IGs, were measured using the above-mentioned NMR techniques. The existence of two regimes of ion dynamics for IGs was observed: a long-range translational diffusion of ions in regions of relatively free IL and restricted diffusion occurring through interactions very close to the silica particles.

■ THEORY

The spin−lattice relaxation of ^{1}H and ^{19}F (spins of quantum number $I = 1/2$) in liquids is mainly due to the intramolecular and intermolecular dipole−dipole interactions, modulated in time by molecular rotations and relative translational diffusion of molecules carrying the nuclear spins, respectively.[18] Considering the chemical structures of the cations and anions in [BMIM] TFSI IL (Figure 1a), it is noted that they contain ^{1}H and ^{19}F, respectively. Therefore, it is important to consider both types of intermolecular interactions in the liquid, the homonuclear (co-ions) and heteronuclear (counter-ions) interactions. As a first approach, the proton and fluorine spin−lattice relaxation dispersions of ILs can be modeled by considering translational diffusion plus a Lorentzian term to account for molecular rotations.

In this work, the translational diffusion was modeled using the so-called force-free-hard-sphere (FFHS) model.[19,20] This model is based on the assumption that the molecular motion

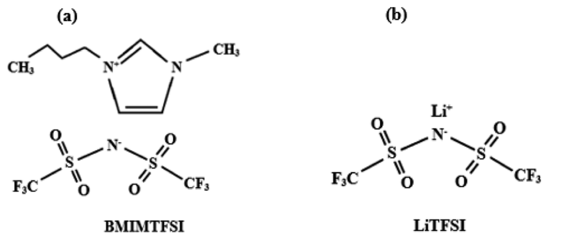

Figure 1. Molecular structure of the components of IL: (a) [BMIM] TFSI and (b) LiTFSI.

can be compared to that of a rigid sphere that obeys Fick's diffusion equation, the spins are positioned in the center of the molecules, and there is a uniform distribution of the molecules outside the distance of the closest approach d. This last one imposes a boundary value condition because of the fact that spin−pair interactions are neglected for interspin distances that are lower than the minimum distance d. The FFHS diffusion model, which includes an excluded volume effect as a consequence of the last assumption, is treated by including an extra term in the relative diffusion equation, representing the potential from averaged forces between the spin-bearing molecules.[19] This model provides an approximate but improved description of the system, as compared with the approach used by other authors,[21] where the excluded volume effect was not considered. In spite of the fact that IL consists of nonspherical ions, the FFHS model has been employed successfully to describe translational diffusion in similar systems.[14,15,22,23] To the best of our knowledge, there are no previous studies that extend the diffusion model for elongated molecules or ions. Therefore, the FFHS model may be considered as a very first approximation to describe the translational diffusion in ILs. Considering that the diffusion can be represented by finite jumps, in the Brownian limit $r^2/6d^2 \to 0$, where r^2 is the mean square jump distance defined by $r^2 \equiv 6D\tau$ and τ is the mean time between jumps, it is possible to arrive at the following expressions for the homonuclear (eq 1) and heteronuclear (eq 2) translational relaxation rates R_1[18,20]

$$R_{1\mathrm{Diff}}^{ii}(\omega_i) = \frac{A_{\mathrm{D}}^{ii}}{d_{ii}D_{ii}}[\tilde{J}(z(\omega_i)) + 4\tilde{J}(z(2\omega_i))] \tag{1}$$

$$R_{1\mathrm{Diff}}^{ij}(\omega_i) = \frac{A_{\mathrm{D}}^{ij}}{d_{ij}D_{ij}}[\tilde{J}(z(\omega_i - \omega_j)) + 3\tilde{J}(z(\omega_i)) + 6\tilde{J}(z(\omega_i + \omega_j))] \tag{2}$$

where

$$A_{\mathrm{D}}^{ii} = \frac{8}{45}\pi\gamma_i^4\hbar^2\left(\frac{\mu_0}{4\pi}\right)^2 n_i \tag{3a}$$

$$A_{\mathrm{D}}^{ij} = \frac{1}{9}\gamma_i^2\gamma_j^2\hbar^2\left(\frac{\mu_0}{4\pi}\right)^2 n_j \tag{3b}$$

$$\tilde{J}(z) = \frac{1 + \frac{5}{8}z + \frac{z^2}{8}}{1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^5}{81} + \frac{z^6}{648}} \tag{4}$$

the subscripts i and j refer to proton and fluorine, respectively, $\omega = 2\pi\nu_0$, and $z(\omega_i) \equiv \sqrt{2\omega_i d_{ij}^{\,2}/D_{ij}}$. D_{ij} is the relative diffusion coefficient defined as the sum of the self-diffusion coefficients

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of the pair of molecules involved $D = D_i + D_j$; d_{ij} is the closest distance between two nuclei located in different molecules, n_i is the nuclear density, $\hbar$ is Plank's constant divided by 2π, and μ_0 is the vacuum magnetic permeability.

On the other hand, isotropic molecular rotations with an average correlation time were assumed as a first approach. Therefore, a Lorentzian form[18,24] can express the contribution of molecular rotations to the spin−lattice relaxation rate

$$R_{1\mathrm{Rot}}^{ii}(\omega_i) = A_\mathrm{R}\left[\frac{\tau_\mathrm{R}}{1 + (\omega_i)^2\tau_\mathrm{R}^{\,2}} + \frac{4\tau_\mathrm{R}}{1 + (2\omega_i)^2\tau_\mathrm{R}^{\,2}}\right] \tag{5}$$

$$A_\mathrm{R} = \frac{3}{10}\gamma_i^4\hbar^2\left(\frac{\mu_0}{4\pi}\right)^2\frac{1}{r_i^{\,6}} \tag{6}$$

where A_R is the corresponding amplitude reflecting the strength of the relevant ^{1}H−^{1}H (^{19}F−^{19}F) dipolar interactions, r_i is the effective ^{1}H−^{1}H (^{19}F−^{19}F) distance between two nuclei located in the same molecule, and τ_R is the average rotational correlation time.

If the two processes described above are assumed statistically independent and/or dominant in different time scales, the total spin−lattice relaxation rate will be given by

$$\begin{aligned} R_{1\mathrm{IL}}^{\;I=1/2}(\omega) &= R_{1\mathrm{intra}}(\omega) + R_{1\mathrm{inter}}(\omega) \\ &= R_{1\mathrm{Rot}}^{ii}(\omega) + R_{1\mathrm{Diff}}^{ii}(\omega) + R_{1\mathrm{Diff}}^{ij}(\omega) \end{aligned} \tag{7}$$

Equation 7 is used to address the measured hydrogen and fluorine relaxation rate dispersions of ILs in this work.

The molecular dynamics becomes more complex when the IL is confined in a silica matrix as it occurs in the IG. Based on the N_2 adsorption isotherm curve (Figure S1) and from the literature,[3] an acid-catalyzed silica IG consists primarily of "inkwell" and some slit shaped pores. The inkwell pores can be thought of as a string of interconnected spherical pores with narrower openings between each one. As such, the IG is two interpenetrating phases: the solid state SiO_2 matrix that has a mesoporous architecture and the IL electrolyte. Some fraction of the IL is close to the SiO_2 particles, and some fraction of the IL is further away. In this case, for simplicity, we assume the coexistence of just those two fractions of the liquid. Previous models have been developed for liquids in porous host materials in which there exists a core fraction, near the pore center and a surface fraction, near the confining walls.[12,15,16,25,26] The first one is dominant, and its ions dynamically behave as "bulk-like". For the second one, the ions interact with the pore walls, and therefore, their dynamics are affected by the matrix surface topology. Specifically, the intramolecular dipole−dipole interaction is modulated in time by translational displacement of the molecules along a rough shaped surface with different local orientations. These molecular reorientations mediated by translational displacements (RMTDs) of the molecules, which occur on a much slower time scale than bulk rotational or translational diffusion, are an efficient relaxation mechanism.[12,25,27,28] We use the basic tenets of these models in our analysis of the data, but because of the morphology of the two interpenetrating phases of the IG described above, for generality, we now identify the bulk-like phase as ions at least a few ionic radii distant from the silica surfaces and the surface-like phase as those ions close to the silica surfaces. Therefore, the total proton and fluorine spin−lattice relaxation rate for IGs will be given by

$$R_{1\mathrm{IG}}(\omega) = W_\mathrm{bulk}R_{1\mathrm{bulk}}(\omega) + W_\mathrm{surf}R_{1\mathrm{RMTD}}(\omega) \tag{8}$$

where W_bulk and W_surf are the weight factors for each corresponding fraction of the liquid, and $R_{1\mathrm{bulk}}$ and $R_{1\mathrm{RMTD}}$ have the following expressions[25,26,28]

$$R_{1\mathrm{bulk}}(\omega) = R_{1\mathrm{IL}}^{\;I=1/2}(\omega) \tag{9}$$

$$\begin{aligned} R_{1\mathrm{RMTD}}(\omega) = A_\mathrm{RMTD}\Bigg[&\omega^{-0.5}\left(f\left(\frac{\omega_\mathrm{max}}{\omega}\right) - f\left(\frac{\omega_\mathrm{min}}{\omega}\right)\right) \\ &+ 4(2\omega)^{-0.5}\left(f\left(\frac{\omega_\mathrm{max}}{2\omega}\right) - f\left(\frac{\omega_\mathrm{min}}{2\omega}\right)\right)\Bigg] \end{aligned} \tag{10}$$

$$\begin{aligned} f(\omega) = {}&\arctan((2\omega)^{0.5} + 1) + \arctan(2\omega)^{0.5} - 1) \\ &- \operatorname{arctanh}\left(\frac{(2\omega)^{0.5}}{\omega + 1}\right) \end{aligned} \tag{11}$$

In eq 8 it should be noted that W_bulk and W_surf have not been considered in prior work. The constant A_RMTD in eq 10 depends on the residual dipole−dipole interaction that is averaged by local molecular orientations, on the microstructural features of the confining matrix, on the diffusion coefficient, and on the fraction of the molecule at the surface. ω_min and ω_max represent, respectively, the minimum and maximum frequencies of diffusion modes on the silica surface and are given by

$$\omega_\mathrm{min} = \frac{4D_\mathrm{surf}}{l_\mathrm{max}^{\;2}} \tag{12}$$

$$\omega_\mathrm{max} = \frac{4D_\mathrm{surf}}{l_\mathrm{min}^{\;2}} \tag{13}$$

where D_surf denotes the surface diffusion coefficient, l_max describes the largest, and l_min describes the smallest displacement distance along the surface. It is important to note that eq 10 is obtained by assuming normal two-dimensional diffusion that can be described by a Gaussian probability density and that the relaxation rate $R_{1\mathrm{RMTD}} \sim \omega^{-0.5}$ behavior in the range between ω_min and ω_max is expected with an assumption of an equipartition of wave numbers describing different diffusion modes (orientational structure factor is constant).[29]

It is worth mentioning that the concept of the bulk and surface fractions assumes that the residence lifetime of the ions on the surface is longer than the translational correlation time in bulk. In this way, the exchange between the two fractions is slow enough to allow us to distinguish between these two populations, and the contribution to the relaxation from the exchange process can be omitted.

The spin−lattice relaxation of nuclei with spin $I \geq 1$, such as ^{7}Li with $I = 3/2$, is mainly due to the interaction between the nuclear electric quadrupole, as characterized by its moment Q, and the fluctuating electric field gradient tensor at the nuclear site, caused by the surrounding intra and intermolecular charge distributions. In the case of our ILs, the origin of the electric field gradient is intermolecular (for a simple Li^+) and its temporal fluctuations include both long-range translational diffusion and local (short-range) dynamics of Li^+. It has been shown[13] that Li^+ can form aggregates with $TFSI^-$ and that the translational motion of Li^+ is given by jumps within the coordination sphere surrounded by anions.[30,31] Therefore, we may consider that the fluctuations of the electric field gradient tensor are mostly due to local translational motion of Li^+. In addition, the fluctuations can be enhanced by the reorientation of the anions. Assuming isotropic diffusional motion with an average correlation time τ_{jump} as a first approach, a Lorentzian form[18,24] can express the spin–lattice relaxation rate, which is in agreement with other works[31,32]

$$R_{1IL}^{Li}(\omega) = A_Q\left[\frac{\tau_{jump}}{1 + (\omega)^2\tau_{jump}{}^2} + \frac{4\tau_{jump}}{1 + (2\omega)^2\tau_{jump}{}^2}\right] \quad (14)$$

$$A_Q = \frac{3\pi^2}{10}\frac{(2I + 3)}{I^2(2I - 1)}\left(1 + \frac{\eta^3}{3}\right)C_Q{}^2 \quad (15)$$

where $\tau_{jump} \equiv \langle r\rangle^2/6D$, $\langle r\rangle^2$ is the average one-jump distance and D is the diffusion coefficient; η is the asymmetry parameter of the electric field gradient and C_Q is the quadrupolar coupling constant. If an axially symmetric electric field gradient tensor is assumed, η is zero. Equation 14 was used to explain the measured lithium relaxation rate dispersions of ILs in this work.

As shown for hydrogen and fluorine, the molecular dynamics becomes more complex when the IL is confined into a silica matrix (IG), but again, it can be treated by assuming the existence of two fractions of the liquid. The same assumption could be made in the case of Li^+ in the IG, and the total spin–lattice relaxation rate for IGs will be again given by

$$R_{1IG}^{Li}(\omega) = W_{bulk}R_{1bulk}(\omega) + W_{surf}R_{1RMTD}(\omega) \quad (16)$$

where R_{1bulk} and R_{1RMTD}[17,29,33−35] have the following expressions

$$R_{1bulk}(\omega) = R_{1IL}^{Li}(\omega) \quad (17)$$

$$R_{1RMTD}(\omega) = A_{RMTD\_power}\omega^{-p}, \qquad 0.5 < p < 1 \quad (18)$$

Equation 18 was obtained by assuming normal two-dimensional diffusion that can be described by a Gaussian probability density and a power law for the orientational structure factor.[29] The amplitude A_{RMTD_power} can be written as[29] $C \times D^{p-1}$, where C is a constant that depends on the type of interaction (quadrupolar in our case), and in particular, on the exponent p, and the temperature dependence is in this case introduced by the diffusion coefficient $D \equiv D_{surf}$. It is interesting to note that the simplest assumption that was made for ^1H and ^{19}F relaxation, namely, a constant orientational structure factor ($p = 0.5$), was not considered in this case. According to many other studies, this assumption works quite well for ^1H and ^{19}F relaxation, but to the best of our knowledge, there are no previous studies using RMTDs to describe ^7Li relaxation. Therefore, a more complex function for the orientational structure factor was chosen.

It is worth mentioning that the molecular RMTDs also produce temporal fluctuations in the electric field gradient; therefore, they are an efficient relaxation mechanism for Li^+ that interact with the silica surface. This has been shown for other quadrupolar nuclei where the relaxation rate at the lower frequencies and the orientational structure factor are given by a power law.[33]

■ EXPERIMENTAL SECTION

Materials and IG Preparation. The IL electrolytes were composed of [BMIM] TFSI with various concentrations of lithium salt bis(trifluoromethanesulfonyl)imide (LiTFSI), namely, 0, 0.5, and 1 M. The molecular structure of [BMIM] TFSI and LiTFSI are shown in Figure 1. The IGs were prepared using methods described in a previous publication in which tetramethylorthosilicate and triethylvinylorthosilicate were hydrolyzed with formic acid.[3] Briefly, the IG matrix was synthesized using a nonaqueous organic acid catalyzed sol–gel process to form a nanoporous silica matrix around an IL, thus confining the electrolyte. The choice of silica precursors and the use of an organic acid allows for a crack free monolith to form under rapid gelation conditions. The silica component maintains a highly porous network with surface areas up to 750 m^2/g and an average pore size of 20 nm as measured through N_2 adsorption. The IL electrolyte component constitutes 75 vol % of the IG. A cumulative pore volume versus pore size plot is provided as calculated from a N_2 adsorption isotherm, shown in Figure S1, using the Micromeritics Tarazona DFT model for cylindrical-pore oxides. Transmission electron microscopy imaging of a similar supercritically-dried IG structure has been provided in previous reports.[3] The image suggests a similar silica particle size distribution as that of the pore size distribution. The sol mixture was aged overnight before it was dried in a vacuum oven. The condensation is assumed to be complete, and the negatively charged surface Si–OH moieties are assumed to be static. The dried solid electrolytes were stored and handled in an argon filled glove box and hermetically sealed in NMR tubes before usage.

NMR PFG Measurements. NMR diffusion experiments were performed with a 300 MHz Direct Digital Drive Varian spectrometer in a magnetic field of 7.05 T with a DOTY Z-gradient diffusion probe. PFG-NMR stimulated echo sequence was used to measure the self-diffusion coefficients of the cations (Li^+, $BMIM^+$) and anion ($TFSI^-$), in the ILs and IGs, at variable temperatures covering the 20–100 °C range. The gradient strength g was arrayed for each experiment (32 or 64 values, linear increase, g = 0–1380 G/cm). The gradient pulse duration was δ = 1–4 ms, and the diffusion delay was Δ = 35–500 ms. The self-diffusion coefficients, D, were then extracted by fitting the decay echo signal with the Stejskal–Tanner equation

$$I = I_0 \exp\left[-D\delta^2 g^2\gamma^2\left(\Delta - \frac{\delta}{3}\right)\right] \quad (19)$$

where I is the amplitude of the attenuated echo signal, I_0 is the initial intensity, and γ is the gyromagnetic ratio.

***R*₁ Measurements.** ^1H, ^{19}F, and ^7Li relaxation rate dispersions for ILs and IGs were measured using a Spinmaster FFC2000 CDC Relaxometer (Stelar; Mede, Italy) in samples of 1 mL volume. The samples consisted of [BMIM] TFSI with 0, 0.5, and 1 M LiTFSI. The spin–lattice relaxation rate R_1 was determined using the standard prepolarized and nonpolarized sequences.[36] In all cases, polarization and acquisition magnetic fields of 0.625 T (equivalent to 25 MHz for ^1H Larmor frequency) and 0.450 T (equivalent to 18 MHz for ^1H Larmor frequency) were used, respectively. Relaxation dispersion curves were measured with 20–25 points distributed in a window ranging from 100 kHz to 35 MHz in ^1H Larmor frequencies (relaxation magnetic field values). A field slew rate of 13 MHz/ms was used in all cases, with a switching time of 2.5 ms. Typically, between 4 and 36 signal scans were accumulated at each delay τ (16 different values) and

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frequency. R_1 was calculated from the magnetization recovery curves that were not sensitive to the time window over which the NMR signal was acquired (after a 90° pulse). The spin relaxation recovery was monoexponential, within acceptable errors, at all frequencies. Relaxation profiles were measured at three different temperatures: 30, 40, and 60 °C. The sample temperature was controlled within ±1 °C using a Stelar variable temperature controller.

Measured relaxation dispersion curves were modeled in terms of a previously discussed approach.[37] The methodology is essentially a least square fitting procedure but with absolute differences between the model and experimental values being minimized. This practice reduces the computing time, while the final result is not critically dependent on this choice. In this method, we averaged the absolute value of deviations between the model curve and the experimental data. We refer to this quantity as SUM, which represents the percentage error of the model curve, in which all fittings were less than 9%. The optimal model fitting to the data was obtained by automatically finding the best combination of model parameters that made SUM lower than a predefined value called SUM_{max} (typically 0.05 for ^1H and ^{19}F and 0.08 for ^7Li). Parameter uncertainties were determined by analyzing the sensitivity of the optimal model curve to variations of each parameter. The error interval associated with each parameter corresponds to the maximum shift of the model curve within experimental errors.

■ RESULTS AND DISCUSSION

Self-Diffusion. Considering the chemical structure of the cations and anions in our samples (Figure 1), studying self-diffusion of ^1H, ^{19}F, and ^7Li is equivalent to studying diffusion of $BMIM^+$, $TFSI^-$, and Li^+, respectively. This was achieved by using PFG-NMR to determine the diffusivity of each ion individually. The self-diffusion coefficients of the ILs and the IG's cations (i.e., $BMIM^+$ and Li^+) and anion (i.e., $TFSI^-$) with different concentrations of LiTFSI are shown in Figures 2 and 3 as a function of temperature. Experimental uncertainties were typically less than 4% and within the size of the data points. It can be seen that these coefficients follow a linear behavior in the Arrhenius plots ($R^2 > 0.99$), and their numerical values are compiled in Tables S1−S3 in Supporting Information. In the case of the ILs where the echo signal decay could not be fitted with a single exponential, weighted average values were calculated for the diffusion coefficients (D_{Avg}) and their values are shown in Figures 2 and 3. For the IGs, the different values exhibited a significantly greater disparity than in the ILs, and thus, their weighting factors are listed separately (Tables S1−S3), and only the bulk component is shown in these two figures. A typical diffusion decay profile demonstrating the inadequacy of a single-value fit is shown in Figure S2.

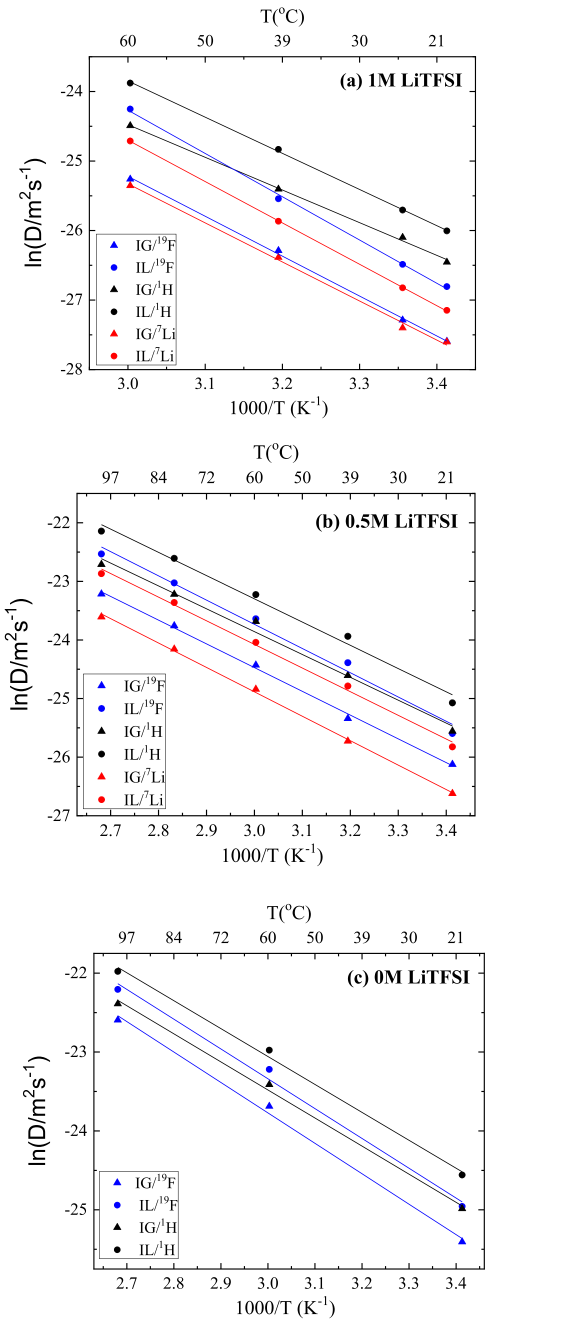

Figure 2. Temperature dependence of the self-diffusion coefficients of Li^+, $TFSI^-$, and $BMIM^+$ in ILs (weighted average values) and IGs (bulk component) at three concentrations of LiTFSI; (a) 1, (b) 0.5, and (c) 0 M.

On the basis of ion sizes,[8] $BMIM^+ > TFSI^- > Li^+$, the Stokes−Einstein equation would predict that the order of the self-diffusion coefficients should be $D_{Li^+} > D_{TFSI^-} > D_{BMIM^+}$. However, in spite of the fact that Li^+ is the smallest ion in the system, it exhibits the smallest diffusion coefficient, followed by $TFSI^-$ and $BMIM^+$. On the other hand, $BMIM^+$, constituting the biggest ion, shows the largest diffusion coefficient over the entire temperature range for all the LiTFSI concentrations (Figure 2). This suggests that $BMIM^+$ might diffuse as a single ion, whereas neither $TFSI^-$ nor Li^+ diffuse as single ions, but rather as Li^+ and $TFSI^-$ aggregates.[8]

In the case of ILs, single-component diffusion is apparent for Li^+ for all the samples, but in the case of $TFSI^-$ and $BMIM^+$, two-component diffusion is observed at some temperatures and concentrations of LiTFSI (Tables S1−S3). Diffusion results obtained for the IG samples display one or two diffusing components, depending on the temperature and the concentration of the LiTFSI. This behavior is observed for all nuclei (^1H, ^7Li, and ^{19}F). It seems that the lower the temperature and/or the higher the LiTFSI concentration, the more probable the ions behave as a two-component diffusion system. Satisfactory fitting (example shown in Figure S2) for the two-component diffusion was obtained by adding another exponential term to the standard Stejskal−Tanner equation

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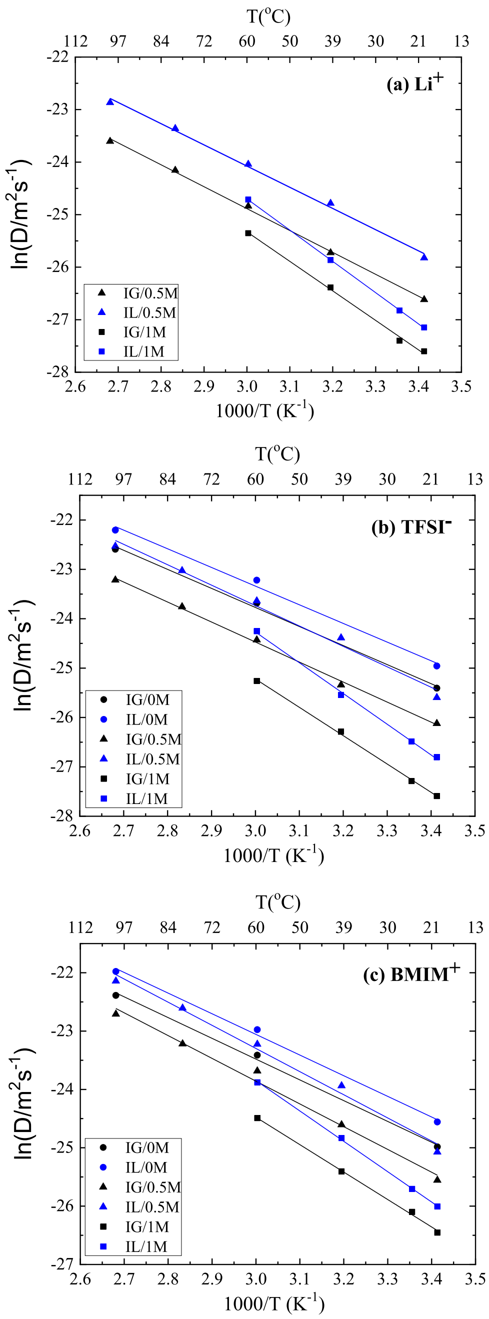

Figure 3. Temperature dependence of the self-diffusion coefficients of (a) Li^+, (b) $TFSI^-$, and (c) $BMIM^+$ in ILs (weighted average values) and IGs (bulk component) at different concentrations of LiTFSI.

$$I = I_0\left[A\ \exp\left[-D_A\delta^2 g^2\gamma^2\left(\Delta - \frac{\delta}{3}\right)\right] + B\ \exp\left[-D_B\delta^2 g^2\gamma^2\left(\Delta - \frac{\delta}{3}\right)\right]\right] \quad (20)$$

where D_A and D_B are the diffusion constants for each diffusion component, and A and B represent the fraction of the carrier content responsible for each diffusion component. For IGs, the weighting factors A and B will be identified as W_{bulk} and W_{surf}, respectively, in the context of the relaxometry results presented later.

Considering the large difference in values obtained for the two diffusion coefficients for each ion species in IGs (Tables S1−S3), the two-component diffusion can be explained by assuming the existence of two distinct populations of ions. The ions located close the confining silica matrix that are impeded by interactions with the silica surface ($B = W_{surf}$) and the ions located far from the silica surface that present a more bulk like behavior with a higher diffusion coefficient ($A = W_{bulk}$). This assumption is consistent with previous works.[6,15] Given typical diffusion timescales (tens of milliseconds), one cannot rule out the exchange between the two species, but we surmise that the charged defects on the silica surface will still maintain a time-averaged population of less mobile ions. On the basis of several trials with different values of Δ resulting essentially in no change of the decay curve (not shown), we do not consider restricted diffusion effects as a possible explanation for the shape of the curve. It is of interest that the dynamics of the ILs appear to be complex as well,[38] showing two-component diffusion that depends on the temperature and the concentration of the lithium salt. Other studies also have identified the existence of two dynamic processes in ILs.[39] From the literature,[8,13] it is known that these ILs can form at least two different kinds of clusters, those composed of $BMIM^+$ and $TFSI^-$ and those composed of Li^+ and $TFSI^-$. Also, several studies[13,40] have proposed that the Li^+ and $TFSI^-$ form $[Li(TFSI)_2]^-$ clusters and even $[Li(TFSI)_4]^{3-}$ clusters for solutions with high contents of LiTFSI. These complex formations between cations and anions could explain the two-component diffusion in ILs,[39] and they are represented by the weighting factors A and B of the eq 20. Though there is undoubtedly some exchange and dynamical averaging between single ions and multiplets, it has been shown by electrophoretic NMR[41] that the multiplets, or at least a time-averaged population of them, diffuse over typical NMR time and length scales. Though much less pronounced for the $BMIM^+$ cations, prior work based on QENS has made a similar observation.[39] However, we emphasize here that the difference between the two D values is much greater in the IG than in the IL.

Arrhenius plots of the diffusion results (Figure 2) show the weighted average value of the diffusion coefficients for ILs and the bulk diffusion values for IGs. It is observed for all three nuclei that the diffusion coefficients increase with temperature and have approximately the same slope (i.e., activation energy) for both 0 and 0.5 M LiTFSI concentrations. For the 1 M LiTFSI sample, the diffusion values increase at a higher rate (i.e., larger activation energy) with increasing temperature compared to the lower LiTFSI concentrations.

From Figure 3, the results show that the diffusion coefficients of the three types of ions are considerably reduced when the concentration of LiTFSI increases. This is primarily because of the increasing viscosity of the IL and corresponding changes in local viscosity in the IG upon the addition of LiTFSI. It is also observed that the confinement causes a decrease by about a factor of two in the diffusion coefficients for $BMIM^+$, Li^+, and $TFSI^-$.[8]

The transport number derived from NMR diffusion is related to the ion transference number, which describes how much charge is carried by the cations or the anions. In order to study the individual contributions of the cations and anions to the overall ionic conductivity of the samples, transport number of the ions can be calculated from the self-diffusion coefficients using the following equation[42,43]

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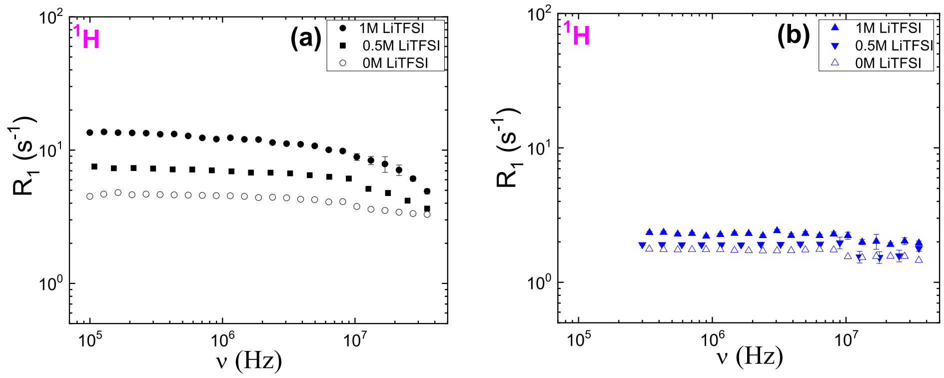

Figure 4. ^{1}H relaxation rate dispersions for [BMIM] TFSI IL with 0, 0.5, and 1 M LiTFSI, recorded at (a) 30 and (b) 60 °C.

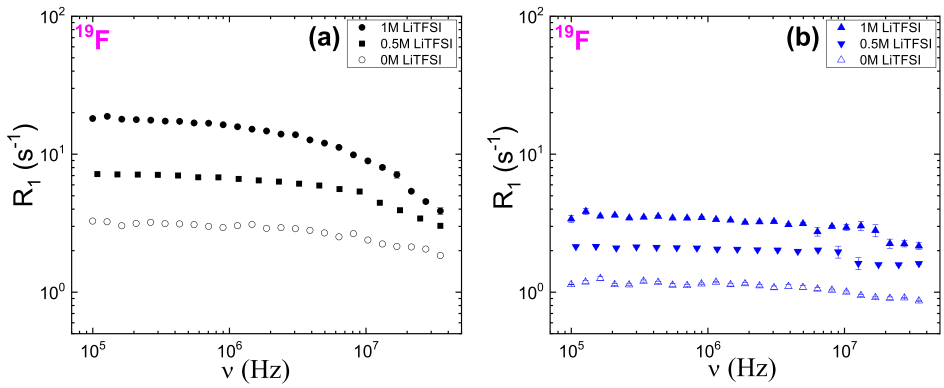

Figure 5. ^{19}F relaxation rate dispersions for [BMIM] TFSI IL with 0, 0.5, and 1 M LiTFSI, recorded at (a) 30 and (b) 60 °C.

$$t_i = \frac{M_i D_i}{\sum_i M_i D_i} \tag{21}$$

where t_i is the transference number, M_i is the ion concentration, and D_i is the self-diffusion coefficients of the ions. The cation transference number receives majority of attention in most electrolyte systems because the charge carrier of interest that stores charge in a battery is most often a cation such as Li. Therefore, in this study, the effective Li-transference number is defined as

$$M_{\mathrm{Li}} \times D_{\mathrm{Li}}/[M_{\mathrm{Li}} \times D_{\mathrm{Li}} + M_{\mathrm{BMIM}} \times D_{\mathrm{BMIM}} + M_{\mathrm{TFSI}} \times D_{\mathrm{TFSI}}] \tag{22}$$

where M_{Li}, M_{BMIM}, and M_{TFSI} are the ion concentrations for Li^+, BMIM^+, and TFSI^- respectively. Interestingly, their values in the liquid electrolyte and in the IG are comparable as shown in Table S4, with a maximum value of 0.1.

Dispersions of the Spin−Lattice Relaxation Rate. The ^{1}H and ^{19}F spin−lattice relaxation rate dispersions for ILs measured at 30 and 60 °C are shown in Figures 4 and 5, respectively. Experimental uncertainties of typically less than 5% are covered by the size of the data points. Both nuclei relax faster when the concentration of lithium salt increases. This is directly connected to the increase in viscosity of the IL upon the addition of LiTFSI and is in agreement with the results obtained from our PFG-NMR diffusion measurements. This effect is stronger for fluorine compared to hydrogen (Figure 5), where variations in the relaxation rate are larger than 100%. This is attributed to the tendency of TFSI^- to interact with Li^+ and form aggregates,[13,40] where the quadrupolar interaction will serve as extra source of relaxation for ^{7}Li which will in turn affect the ^{19}F relaxation through dipolar coupling. The behavior at 60 °C (Figures 4b and 5b) is consistent with decreased viscosity at that temperature (flat relaxation profiles).

If ^{1}H and ^{19}F relaxation profiles for ILs are compared at the same temperature, a "flipping" effect is observed when the concentration of lithium salt increases, as shown in Figure 6 at 30 °C (same effect is observed at 60 °C). For the IL without lithium salt, ^{1}H relaxes faster than ^{19}F, but when the concentration of LiTFSI is 1 M, the opposite occurs. Again, this effect is explained by the strong tendency of TFSI^- to interact with Li^+ and form aggregates.

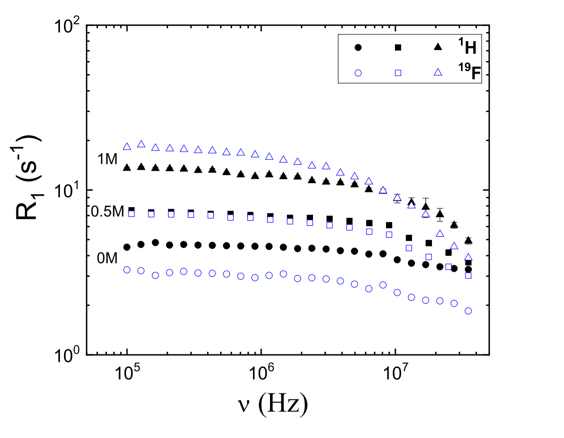

Figure 6. ^{1}H and ^{19}F relaxation rate dispersions for [BMIM] TFSI IL with 0 M (solid and hollow circles), 0.5 M (solid and hollow squares), and 1 M (solid and hollow triangles) LiTFSI, recorded at 30 °C.

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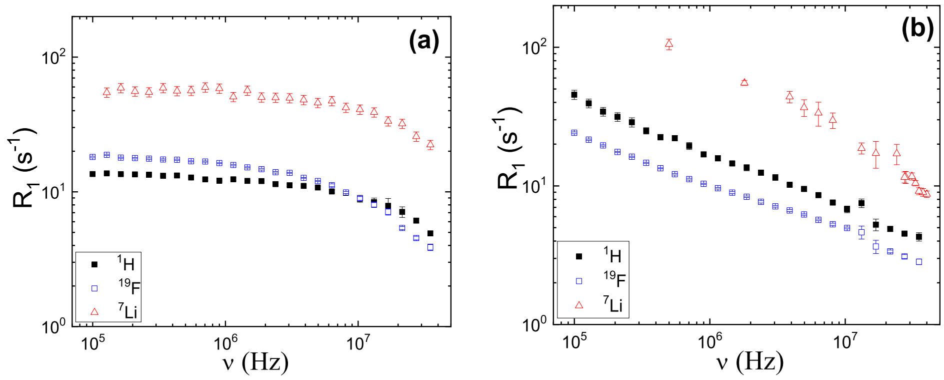

Figure 7. ^{1}H, ^{19}F, and ^{7}Li relaxation rate dispersions recorded at 30 °C, with 1 M LiTFSI. (a) IL and (b) IG.

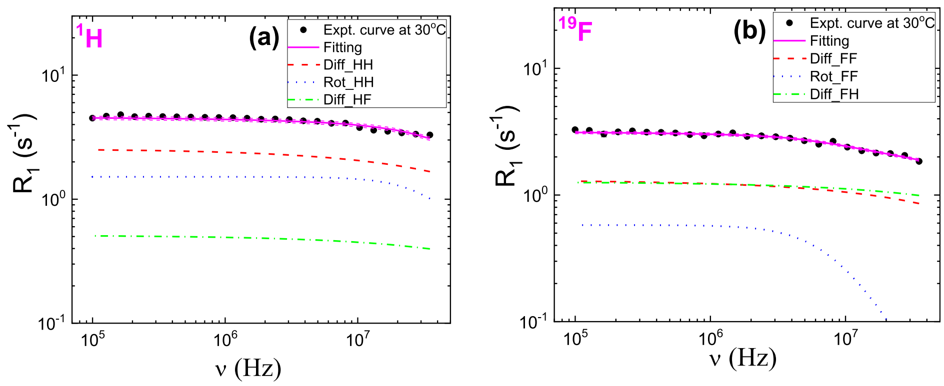

Figure 8. (a) ^{1}H and (b) ^{19}F relaxation rate dispersions for [BMIM] TFSI IL with 0 M LiTFSI and their corresponding fittings using the model given by eq 7, recorded at 30 °C.

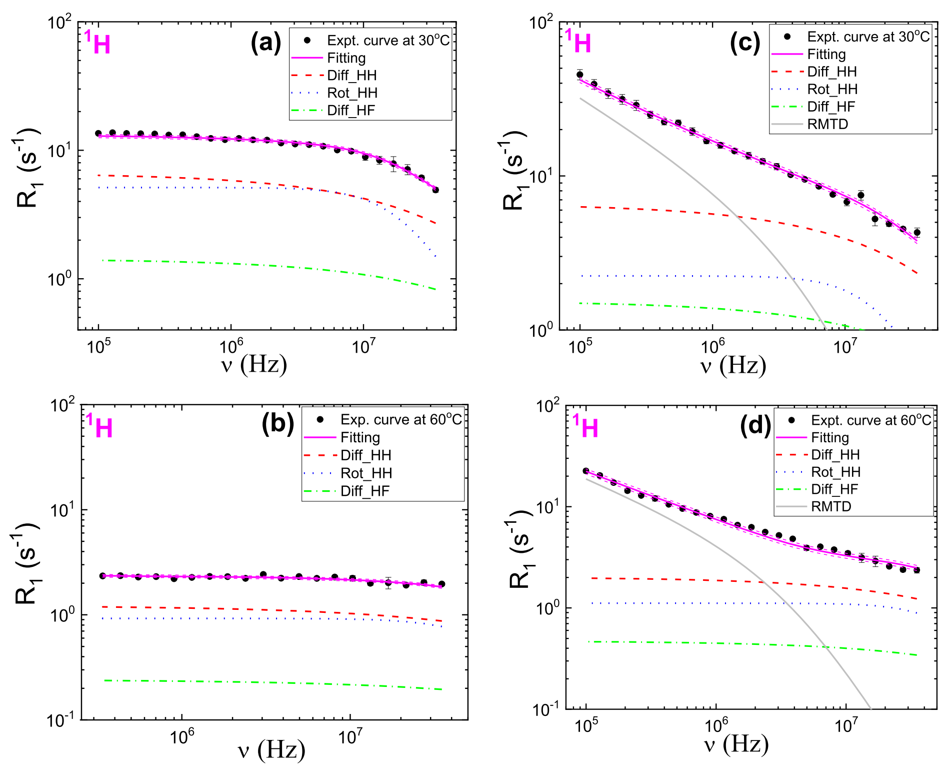

Figure 9. ^{1}H relaxation rate dispersions for ILs and IGs with 1 M LiTFSI and their corresponding fittings using the models given by eqs 7 and 8, recorded at different temperatures. (a) IL at 30 °C, (b) IL at 60 °C, (c) IG at 30 °C, and (d) IG at 60 °C.

The ^{1}H, ^{19}F, and ^{7}Li spin–lattice relaxation rate dispersions for ILs and IGs with 1 M LiTFSI measured at 30 °C are displayed in Figure 7. Experimental uncertainties are either within the size of the data points (<5%) or indicated by error

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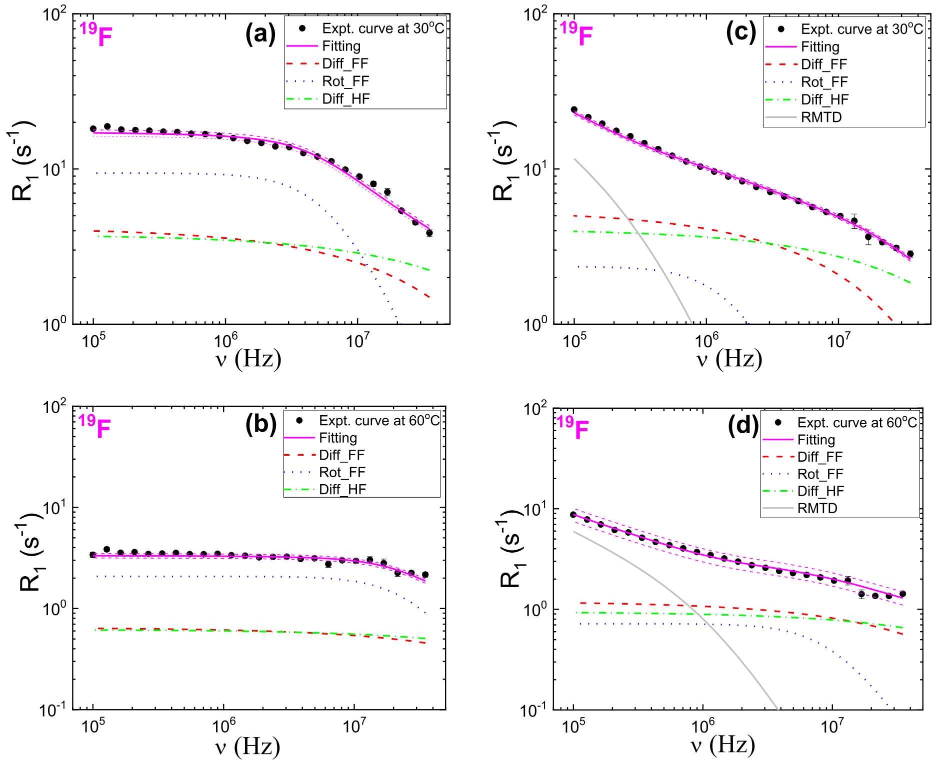

Figure 10. ^{19}F relaxation rate dispersions for ILs and IGs with 1 M LiTFSI and their corresponding fittings using the models given by eqs 7 and 8, recorded at different temperatures. (a) IL at 30 °C, (b) IL at 60 °C, (c) IG at 30 °C, and (d) IG at 60 °C.

bars (up to 20%). Again, considering the chemical structure of the cations and anions in our samples (Figure 1), studying the relaxation rate of ^1H, ^{19}F, and ^7Li is equivalent to studying molecular dynamics of $BMIM^+$, $TFSI^-$, and Li^+, respectively.

From Figure 7a,b, it is observed that the relaxation profiles of the three nuclei present similar behavior, but ^7Li relaxes faster because of its quadrupolar nature. The high efficiency of the quadrupolar relaxation is related to large strength of the interaction: the quadrupole coupling constant for ^7Li is typically an order of magnitude larger than the dipolar coupling constants. For ILs, the relaxation profiles show a plateau at the lowest frequencies, whereas for IGs, they show a monotonic increase when frequencies decrease. This different behavior is a clear indication that a new relaxation mechanism is activated in the IG because of confinement of the IL, namely, molecular RMTDs as will be explained later.

Analysis of Hydrogen and Fluorine Relaxation Profiles. The models given by eqs 7 and 8 were used to analyze the ^1H and ^{19}F relaxation profiles for ILs (0 and 1 M LiTFSI) and IGs (1 M LiTFSI), respectively. All the physical parameters involved in these models were estimated from data fitting by means of fixing them within their most probable intervals obtained from the literature. In particular, the ranges for diffusion coefficients and the weighting factors W_{bulk} and W_{surf} (eq 8) were taken from our own PFG-NMR diffusion measurements (Tables S1–S3). The proton and fluorine densities n (eqs 3a and 3b) were estimated using the corresponding values of molecular weight of each ion and the density of the IL, obtaining $n_H = 3.1 \times 10^{28}$ m^{-3} and $n_F = 1.2 \times 10^{28}$ m^{-3}, for the sample with 0 M LiTFSI and $n_H = 1.8 \times 10^{28}$ m^{-3} and $n_F = 7.2 \times 10^{27}$ m^{-3}, for the sample with 1 M LiTFSI. Using them in eqs 3a and 3b, A_D took the following values: $A_D^{HH} = 9.83 \times 10^{-21}$ m^3/s^2, $A_D^{HF} = 6.90 \times 10^{-22}$ m^3/s^2, $A_D^{FF} = 3.1 \times 10^{-21}$ m^3/s^2, and $A_D^{FH} = 1.73 \times 10^{-21}$ m^3/s^2, for the sample with 0 M LiTFSI. On the other hand, $A_D^{HH} = 5.74 \times 10^{-21}$ m^3/s^2, $A_D^{HF} = 4.04 \times 10^{-22}$ m^3/s^2, $A_D^{FF} = 1.8 \times 10^{-21}$ m^3/s^2, and $A_D^{FH} = 1.01 \times 10^{-21}$ m^3/s^2, for the sample with 1 M LiTFSI.

The obtained fitting curves for ILs with 0 M LiTFSI at 30 °C are shown in Figure 8 and for ILs and IGs with 1 M LiTFSI at 30 and 60 °C are shown in Figures 9 and 10, where good agreement with data is observed within the experimental uncertainties. Curves for other temperatures are shown in Figures S3–S6 in Supporting Information. The corresponding fitting parameters are presented in Tables S5 and S6. Their numerical values follow the expected tendency with increasing temperature, and they are within the expected range according to the literature.[12,15,16,22,23,26]

From Figures 8a and 9, we learn that ^1H relaxation for ILs (without and with lithium salt) is dominated by homonuclear translational diffusion in the whole frequency range. In addition, it is worth noting that while the rotational contribution is quite important in the same frequency range, the heteronuclear translational diffusion contribution is very small because of the lower density of fluorine compared to hydrogen (6 vs 15 nuclei per ion pair). On the other hand, the ^1H relaxation for IGs is dominated by RMTDs in the low-frequency region, namely, $\nu_o < 2$ MHz, and for higher frequencies, the same situation described above for ILs is observed.

If we consider ^{19}F relaxation profiles for ILs without lithium salt (Figure 8b), it can be seen that the relaxation is dominated by homonuclear and heteronuclear translational diffusion in the whole frequency range, and this is in agreement with other works.[22] It is interesting to note that in this case, the heteronuclear translational diffusion contribution is compara-

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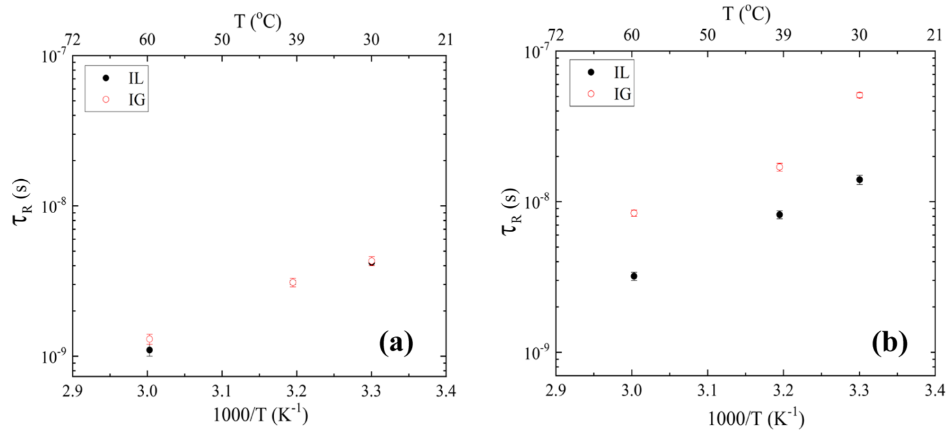

Figure 11. Rotational correlation time of (a) $BMIM^+$ and (b) $TFSI^-$ in ILs and IGs (bulk component) with 1 M LiTFSI at different temperatures.

ble to the homonuclear one. On the other hand, if we analyze ^{19}F relaxation profiles for ILs with lithium salt (Figure 10a,b), it is observed that the relaxation is dominated by molecular rotations over almost the whole frequency range. This unusual feature might be explained by the fact that the translational mobility of $TFSI^-$ is coupled to its interconversion between trans and cis conformers,[44] and this change of conformation makes molecular rotations a very efficient relaxation mechanism. In addition, there is less translational diffusion not only because of the higher viscosity but also due to the closely related aggregation with the lithium ions, which makes intramolecular relaxation dominant. On the other hand, ^{19}F relaxation for IGs (Figure 10c,d) is dominated by RMTDs in the low-frequency region ($\nu_o \leq 600$ kHz), and for larger frequencies, the translational diffusion contributions become dominant.

Regarding ions in ILs and in the bulk phase in IGs, decreasing rotational correlation times are observed when temperature increases (Figure 11), which is consistent with a faster molecular mobility. In addition, it is observed that the confinement has only a minor effect on the molecular rotations of cations which is in agreement with other works,[11,14] although one of these works[14] was for a larger anion. In the case of $TFSI^-$, it was found that the cis−trans conformational transition is inhibited under hydrostatic pressure.[44] We hypothesize that confinement may have a similar effect and that is why we observe that the molecular rotations of anions become slower. Additionally, it can be seen that rotational diffusion is faster for cations compared to anions (Figure 11a,b), at all the temperatures. This is partly due, again, to the tendency of $TFSI^-$ to interact with Li^+ and form aggregates.[13,40]

On the other hand, for ions interacting with the silica surface in IGs, the surface diffusion coefficient (eqs 12 and 13) shows a significant rise when the temperature increases but only for fluorine (Figure 12). This behavior is consistent with the fact that RMTDs seem to be an efficient relaxation mechanism at much lower frequencies for fluorine (~kHz) compared to hydrogen (~MHz) at 30 °C, in other words, $D^F_{surf} < D^H_{surf}$. When the temperature increases, the contribution of this process to the total fluorine relaxation rate becomes comparable to hydrogen relaxation in the same range of frequencies (Figures 9c,d and 10c,d). However, it should be noted that the W_{surf} is typically a factor of two or more larger for ^1H than ^{19}F (Table S3), suggesting a preferential

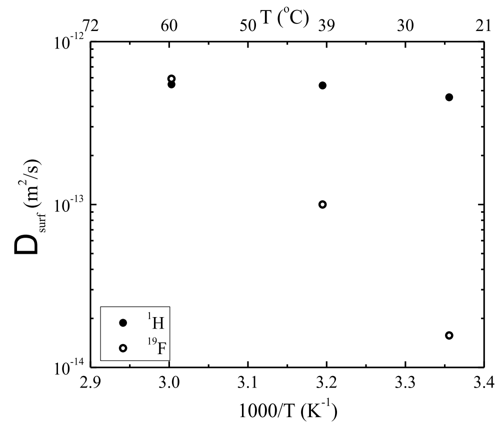

Figure 12. Surface diffusion coefficients of $BMIM^+$ (hydrogen) and $TFSI^-$ (fluorine) in IGs with 1 M LiTFSI at different temperatures, obtained from PFG-NMR measurements. Data taken from Table S2 (Supporting Information) and plotted here to better illustrate different temperature dependencies of BMIM and TFSI. Experimental uncertainties of less than 4% are covered by the size of the data points.

interaction of cations with the silica surface, which in turn, is consistent with the presence of negatively charged defects, presumably hydroxyl groups, in the confining matrix.[9] Therefore, we hypothesize that fluorine interacts indirectly with the silica surface by aggregation with either $BMIM^+$ or Li^+.

Additionally, the hypothesis of the last paragraph can be supported by analyzing the correlation between W_{surf} and Li^+ concentration in IGs. It is observed that $TFSI^-$ and $BMIM^+$ interact with each other mostly in the bulk in the absence of lithium salt (0 M LiTFSI); this means W_{surf} is zero or insignificant for both ions (Table S1). When 0.5 M LiTFSI is added, it seems that $TFSI^-$ remains in the bulk forming aggregates with Li^+ predominantly and interacts with a smaller fraction of $BMIM^+$. This is in agreement with the fact that $BMIM^+$ exhibits a significant fraction that interacts with the silica surface (Table S2). When even more lithium salt is added (1 M LiTFSI), Li^+ starts interacting with the surface and indirectly $TFSI^-$ as well, but still the majority (>75%) of these aggregates remain in the bulk phase (Table S3). In conclusion, only when the concentration of lithium salt is high enough (1 M LiTFSI in our case), a fraction of $TFSI^-$ interacts with the surface.

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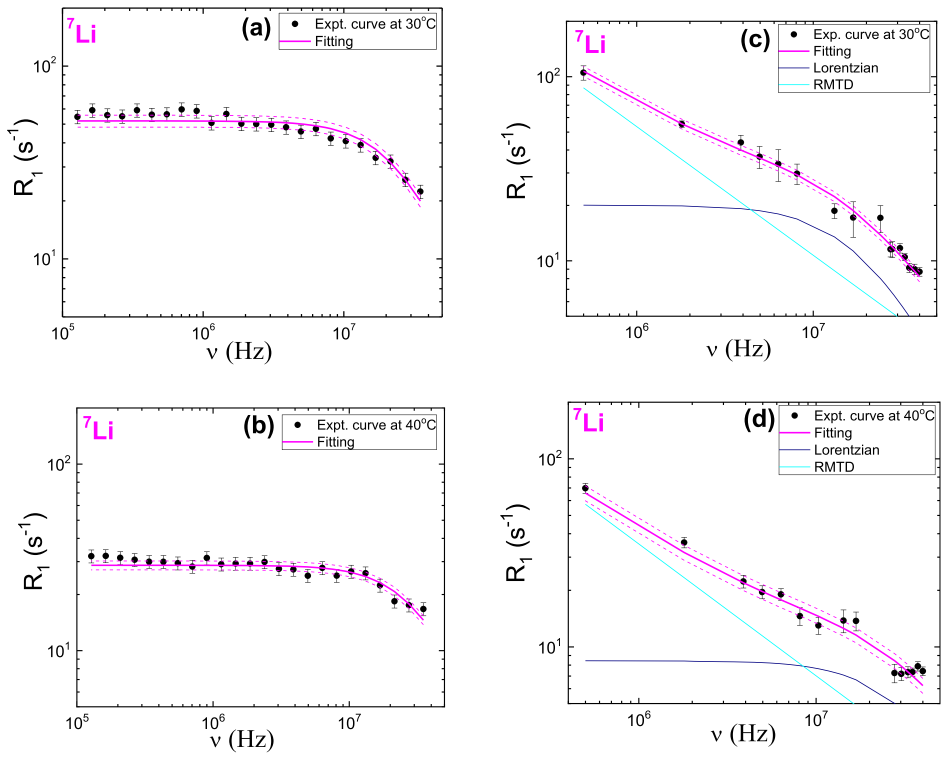

Figure 13. ^{7}Li relaxation rate dispersions for ILs and IGs with 1 M LiTFSI and their corresponding fittings using the models given by eqs 14 and 16, recorded at different temperatures. (a) IL at 30 °C, (b) IL at 40 °C, (c) IG at 30 °C, and (d) IG at 40 °C.

Analysis of Lithium Relaxation Profiles. The models given by eqs 14 and 16 were used to analyze the ^{7}Li relaxation profiles for ILs and IGs with 1 M LiTFSI, respectively. All the involved physical parameters in these models (τ_{jump}, A_Q and D_{surf}) were estimated from data fitting by means of fixing them within their most probable intervals obtained either from literature or from our own PFG-NMR diffusion measurements. In particular, the ranges for τ_{jump} were calculated by using the diffusion coefficients D_{Li1} from Table S3 (assuming that they correspond to short-range diffusion) and considering $\langle r \rangle^2$ smaller than the coordination sphere radius.[31] In addition, the weighting factors W_{bulk} and W_{surf} (eq 16) were taken from the same table.

The obtained fitting curves for ILs and IGs with 1 M LiTFSI are shown in Figure 13, where good agreement with data is observed within the experimental uncertainties. The ^{7}Li relaxation rate dispersion for IL with 1 M LiTFSI recorded at 60 °C is presented in Figure S7 (IG at 60 °C was not measured). Additionally, the corresponding fitting parameters are presented in Table S7. Their numerical values follow the expected tendency with increasing temperature, and they are within the expected range according to the literature.[17,31,32]

Comparison of Figure 13a,c shows that a monotonic increase of lithium relaxation rate is observed for the IG when frequencies decrease. This behavior supports the supposition that a new relaxation mechanism, namely, RMTDs, is activated in the IG because of the interaction of the ions with the silica surface and is very efficient on the slower time scale (even more efficient than bulk translational diffusion). Additional arguments are the fact that two population of ions were observed through the PFG diffusion measurements (Table S3), and the silica surface is negatively charged which makes the interaction with positive ions more probable.[9] From Figure 13c,d, it is observed that the relaxation for IGs is dominated by RMTDs in the low-frequency region, namely, ν_o smaller than few MHz, and for larger frequencies, diffusional motion describes mostly the total relaxation rate similarly to the ILs (Figure 13a,b).

Regarding Li^+ in ILs and in the bulk phase in IGs, decreasing average correlation times τ_{jump} for diffusional motion are observed when temperature increases (Figure 14) which is consistent with a faster molecular mobility. Additionally, it is noted that the confinement makes the diffusional motion of Li^+ slightly slower. A decreasing tendency in the values of A_Q is obtained when temperature increases (Table S7), which means that the quadrupolar coupling becomes weaker at higher temperatures. This is attributed to the temperature-dependent Brownian motion of the nucleus (also the range of different

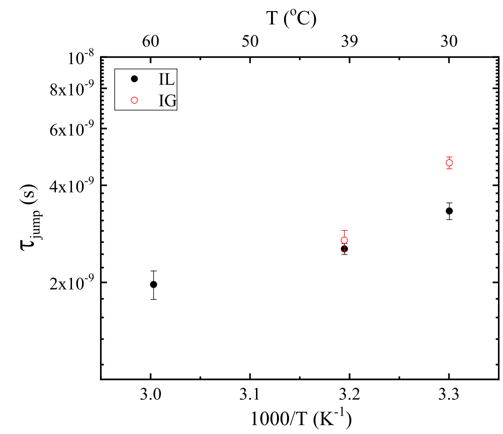

Figure 14. Average correlation time τ_{jump} for diffusional motion of Li^+ in ILs and IGs (bulk component) with 1 M LiTFSI, at different temperatures.

coordination environments), which results in a time average of the electric field gradient.[45]

On the other hand, for the ions interacting with the silica surface in IGs, and which follow eq 18 , where the amplitude A_{RMTD_power} can be written as[29] $C \times (D_{surf})^{p-1}$, the following ratio should be true

$$C_1 \equiv \frac{A^{30\,°C}_{RMTD\_power}}{A^{40\,°C}_{RMTD\_power}} \approx \frac{{D_{surf}}^{n-1}(30\ °C)}{{D_{surf}}^{n-1}(40\ °C)} \equiv C_2 \tag{23}$$

where the amplitude A_{RMTD_power} is a parameter calculated from the analysis of the relaxation profiles, the exponent p is the same at 30 and 40 °C (Table S7 in Supporting Information), and D_{surf} is measured using PFG-NMR. Here, we assume for simplicity that the constant C is approximately the same at both temperatures, although there is likely a small temperature dependence of the quadrupole coupling constant. After doing the corresponding calculations, the values of C_1 and C_2 were 1.08 and 1.13, respectively, which is a clear indication of the consistency between our lithium PFG diffusion and FFC measurements.

CONCLUSIONS

In this work, PFG-NMR and FFC-NMR relaxometry techniques have been combined to study molecular dynamics of the cations and anions in ILs and IGs composed of [BMIM] TFSI with different concentrations of LiTFSI and at different temperatures.

The diffusion coefficients obtained from PFG-NMR diffusion measurements provided an input used to fit the FFC data successfully, which validates the models used to describe the relaxation profiles.

The dynamics of bulk ions slowed down modestly under confinement as shown by the values of the diffusion coefficients (range for ILs: 2×10^{-12} to 3×10^{-10} m^2/s; range for IGs: 1×10^{-12} to 2×10^{-10} m^2/s) and rotational correlation times (range for ILs: 1×10^{-9} to 1×10^{-8} s; range for IGs: 1×10^{-9} to 5×10^{-8} s). In particular, we obtained Li-transference numbers that are comparable for ILs and IGs, which is a clear evidence, that encapsulation of the IL in the SiO_2 matrix does not severely hinder Li^+ transport (despite the observation of some interaction with the silica surface). Therefore, under suitable conditions, properties of IGs approach the properties of ILs, and this behavior is an encouraging indication for using IGs as a solid electrolyte for Li^+ batteries.

The study of ionic transport of Li^+ in [BMIM] TFSI ILs with various concentrations of LiTFSI has established the following important points. (i) Two-component diffusion was observed and explained by assuming the existence of two populations of ions originated by the complex aggregations between cations and anions. (ii) In the case of IGs, the existence of two regimes of ion dynamics was observed as well: a long-range translational diffusion and a restricted diffusion occurring near the surface of the silica particles.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.0c02755.

Cumulative pore volume versus pore size plot calculated from a N_2 adsorption isotherm, values of all measured self-diffusion coefficients and Li-transference numbers, and supplementary figures for relaxation rate profiles with their corresponding fittings and values of fitting parameters (PDF)

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Notes

The authors declare no competing financial interest.

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